



Dimethyltryptamine (DMT): Subjective effects and patterns of use among Australian recreational users

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ABSTRACT

Dimethyltryptamine (DMT) is an endogenous hallucinogen with traditional use as a sacrament in the orally active preparation of ayahuasca. Although the religious use of ayahuasca has been examined extensively, very little is known about the recreational use of DMT. In this study, Australian participants ($n = 121$) reporting at least one lifetime use of DMT completed an online questionnaire recording patterns of use, subjective effects and attitudes towards their DMT use. Smoking DMT was by far the most common route of administration (98.3%) with a comparatively smaller proportion reporting use of ayahuasca (30.6%). The reasons for first trying DMT were out of a general interest in hallucinogenic drugs (46.6%) or curiosity about DMT's effects (41.7%), while almost one-third (31.1%) cited possible psychotherapeutic benefits of the drug. An increase in psychospiritual insight was the most commonly reported positive effect of both smoked DMT (75.5%) and ayahuasca (46.7%), a finding that is consistent with other studies examining the ritualised use of ayahuasca in a religious context. Although previous studies of DMT use have examined ayahuasca use exclusively, the present study demonstrates the ubiquity of smoking as the most prevalent route of administration among recreational DMT users.

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1. Introduction

Dimethyltryptamine (DMT) is a naturally occurring hallucinogen similar in structure to both serotonin and psilocybin. Like other hallucinogens, DMT is a serotonin 5-HT_{2A} receptor agonist (Smith et al., 1998), in addition to being a ligand of both trace amine-associated receptor 1 (TAAR1; Bunzow et al., 2001) and the sigma-1 (σ -1) receptor (Fontanilla et al., 2009). It is found ubiquitously across the plant and animal kingdoms (McKenna, 2004) and, although it is a relatively obscure drug, its presence in the popular media (Otis, 2009; De Conceicao, 2009; Grigoriadis, 2006; Cox, 2009) suggests that interest or use in Western countries is increasing. Szára (1956) was the first to report its hallucinogenic effects, however, its consumption in the hallucinogenic plant beverage ayahuasca dates to pre-Colombian times and continues to the present day (Pomilio et al., 1999; Grob et al., 1996).

Ayahuasca, which translates to “the vine of the souls” by the Amazon-dwelling Quechua people among whom it is traditionally used, is consumed as a sacrament throughout the Amazon Basin by indigenous populations (Dobkin de Rios, 1972; McKenna, 2006). Although it is an umbrella term used to describe any orally active DMT preparation, ayahuasca is usually prepared by infus-

ing the DMT-bearing leaves of *Psychotria viridis* with the stems of the *Banisteriopsis caapi* vine (McKenna, 2004). The latter contains β -carboline alkaloids, chiefly harmine, tetrahydroharmine, and harmaline. Ordinarily, DMT is rapidly metabolised by gut and liver monoamine oxidase (MAO), making it orally inactive. However, the β -carbolines display MAO inhibition and their co-administration enables DMT to reach the CNS intact. The hallucinogenic effects of ayahuasca commence within an hour of its consumption and last approximately 4 h.

In contrast to oral ingestion, the effects of smoked DMT commence almost immediately, peak within several minutes and typically resolve within 30 min (Turner, 1994; Strassman, 2001). Smokeable freebase DMT may be obtained from a variety of flora through a simple extraction process, the instructions for which are readily available on the Internet (Halpern and Pope, 2001). Owing to the brevity and intensity of its effects, smoked DMT has been facetiously coined the “businessman's lunch trip” (Turner, 1994) and this contrasts the duration of LSD, whose effects last 8–12 h (Rothlin, 1957).

In the Amazon Basin, DMT is legal and its use, in the form of ayahuasca, is a key component of the religious practices of several syncretic churches including the União do Vegetal (UDV) and Santo Daime (MacRae, 1998). In the Western world, however, DMT remains a controlled substance, although recent rulings by the United States Supreme Court now protect the religious use of ayahuasca in the United States under the Religious Freedom

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Restoration Act (1993; Bullis, 2008). To date, several studies have examined the religious use and effects of ayahuasca among adolescent (Da Silveira et al., 2005; Doering-Silveira et al., 2005a,b), as well as first-time (Trichter et al., 2009) and long-term (Grob et al., 1996; Halpern et al., 2008) ayahuasca users. These studies have generally found ayahuasca to be psychologically beneficial, or at worst, lacking in deleterious effects when consumed in an appropriate religious context. Users generally report that ayahuasca facilitates an experience characterised by complex and semantically rich visual hallucinations of personal or spiritual significance (Shanon, 2002) and this is consistent with findings that the chemically related hallucinogen psilocybin may evoke religious experiences in hallucinogen-naïve subjects (Griffiths et al., 2006, 2008).

In recent years there appears to have been growing interest in hallucinogens such as DMT in countries such as Australia and the United States (Otis, 2009; De Conceicao, 2009; Grigoriadis, 2006; Cox, 2009). This may be due to the emergence of the outdoor rave or 'doof' music subculture (Luckman, 2003; Tramacchi, 2000) and the proliferation of neo-shamanic practises in the West (Tupper, 2008, 2009). There is an apparent increase in Internet resources surrounding the use of hallucinogens (Halpern and Pope, 2001; Boyer et al., 2005) as well as the expansion of online "head shops" legally trading in the sale of psychoactive plant material, including ayahuasca ingredients (Dalgarno, 2008). Interest may exist in particular for DMT given its endogenous presence in humans (Barker et al., 1981) and its unique and bizarre phenomenology (Shanon, 2002; Strassman, 2001).

Despite the aforementioned studies of ayahuasca, recreational¹ use of DMT in Western countries remains largely unexamined. Previous research has investigated the recreational use of the short-acting hallucinogen *Salvia divinorum* (Gonzalez et al., 2006; Lange et al., 2008), yet a similar study of DMT has not previously been undertaken. In his ethnographic study of Australian DMT users, Tramacchi (2006) examined the phenomenological aspects of the DMT experience, however, little is known regarding the patterns of DMT use among recreational users. Furthermore, data regarding the use of smokeable DMT remain scant.

The present study seeks to address the paucity of such data by examining Australian recreational DMT users. The use of DMT within Australia may be of particular interest given the presence of DMT in several native species of *Acacia* (Fitzgerald and Sioumis, 1965; Rouvelli and Vaughan, 1967) as well as the veritable existence of an electronic 'bush doof' music subculture with which hallucinogen use is commonly associated (Tramacchi, 2000; Luckman, 2003). In the absence of existing literature regarding recreational DMT use, this study is exploratory in nature and seeks to identify the demographics and general pattern of use, as well as subjective effects and attitudes towards DMT in a sample of recreational users of the drug.

2. Method

2.1. Sample

Individuals who were Australian residents aged 18 years and over and who had used DMT at least once in their lifetime were recruited for the study. Data were collected between July and August 2009, with participants recruited via 'snowballing' (Biernacki and Waldorf, 1981) involving individuals known to the researchers, and an advertisement in an Australian hallucinogen-related internet newsletter that contained a link to an online questionnaire. Respondents were assured of their anonymity and the study conformed to National Health and Medical Research Council ethics guidelines for human research.

¹ Given its traditional use as a sacrament it is likely that DMT use deemed to be 'recreational' involves the use of this drug for religious purposes. Although it is acknowledged as a misnomer, in the present study the term 'recreational' will imply the illicit use of DMT outside of a recognised religious institution such as Santo Daime.

2.2. Questionnaire

The self-administered online questionnaire included both closed and open-ended items and took approximately 25 min to complete. The questionnaire comprised four main sections: (1) demographic information: including age, gender, postcode, religious affiliation, education and occupation; (2) history of other drug use: including previous drugs consumed, and drugs consumed in the previous 12 months, 6 months, and 30 days; (3) history of DMT use: including age of initiation, number of occasions used, routes of administration, sources for obtaining DMT, and locations consumed; (4) patterns of use and attitudes about DMT: including reasons for first consuming DMT, positive and negative aspects of both smoked DMT and ayahuasca, patterns of use of each route of administration, and perceived risks and harm reduction regarding DMT use.

Items regarding DMT use were distinguished via the two main routes of administration, smoked and ayahuasca. Respondents were asked to complete sections relevant to their reported route of administration; those who had consumed DMT via both routes were required to complete both sections. Additionally, cannabis dependence was measured using the Severity of Dependence Scale (SDS) adapted for cannabis use in the previous 30 days. The SDS is a 5-item questionnaire with reliability in identifying cannabis dependence (Hildes et al., 2007; Martin et al., 2005). Although a cut-off SDS score of ≥ 3 is considered optimal (Swift et al., 1998), the present study used a cut-off of ≥ 4 in order to increase measure specificity.

2.3. Data analyses

The majority of the analyses were descriptive in nature. Percentages were reported for categorical variables. For continuous normally distributed variables, mean, standard deviation and range were reported and *t*-tests were employed with a 0.05 level of significance. For skewed continuous variables, median and range were reported and the Mann-Whitney *U*-test was used. Qualitative responses for open-ended questions were coded according to themes and percentages reported. Unless otherwise noted, the response rate of each item was $n = 121$.

3. Results

3.1. Demographic characteristics

A total of 121 lifetime DMT users were recruited, the majority of whom were male (86%). Respondents ranged in age from 18 to 68 years, with a median age of 28 years. The majority of respondents were Australian-born (86.8%) and one reported being of indigenous Australian descent. Almost three-quarters (73.6%) reported having no religious affiliation, and only 4.1% belonged to a mainstream monotheistic religion such as Christianity. The remainder cited pantheistic religions (e.g. shamanism; 11.6%), esoteric or non-specific beliefs (e.g. "non-religious, intuitively spiritual"; 8.3%), or Eastern mysticism (e.g. Buddhism; 2.5%). Mean of total years of formal education was 15.9 (S.D. 2.9; range 10–24), with 60.3% of the sample having completed some university or higher. One-third (35.5%) reported full-time employment, with another third reporting part-time or casual employment (34.8%), 14.8% identifying themselves as students and 10.7% as unemployed. Another 4.1% of the sample listed home duties as their primary occupation.

3.2. Overall pattern of drug use

Lifetime history of drug use was extensive among the sample (Table 1). A total of 92 discrete drugs had been used by the respondents. The mean number of drugs ever tried was 17.3 (maximum allowed 35, range 4–34.). Although the use of other hallucinogens was typical, drug use extended to a disparate array of drug classes including stimulants, opiates, depressants and anaesthetics. Use of various mildly psychoactive plants such as damiana (*Turnera diffusa*) was also reported but excluded from this analysis.

All but two respondents (98.3%) reported lifetime use of alcohol and cannabis, and 89.3% and 73.6% had consumed these drugs in the previous 30 days, respectively. Almost one-third of the sample (31.4%) had consumed five or more standard drinks of alcohol in a single sitting on five or more occasions in the past 30 days. Using a cut-off score of ≥ 4 in the SDS for cannabis use in the previous 30 days, 26.4% of the sample appeared to be cannabis dependent.

Table 1
Patterns of drug use of DMT users in the study ($n = 121$).

Drug class	Ever used (%)	Used last 6 months (%)	Used last 30 days (%)	Days used last 30 days (median) ^a
DMT	100.0	68.6	34.7	2
Cannabis	98.3	82.6	73.6	15
Alcohol	98.3	91.7	89.3	8
LSD	96.7	63.6	30.6	1
Tobacco	95.9	74.4	66.9	29
MDMA	94.2	55.4	31.4	1
Psilocybin	91.7	50.4	29.8	1.5
Meth/amphetamine	83.5	26.4	14.0	3
Nitrous oxide	79.3	33.9	14.9	1
Cocaine	72.7	24.8	12.4	1
<i>Salvia divinorum</i>	67.8	14.9	6.6	1
Mescaline	64.5	21.5	8.3	1
Amyl nitrite	60.3	8.3	5.0	1.5
Benzodiazepines ^b	58.7	18.2	9.9	2.5
Ketamine	56.2	14.0	5.0	1
Other opiates ^b	50.4	12.4	8.3	2.5
2C-B	44.6	17.4	4.1	1
MDA	43.0	2.5	1.7	2
<i>Peganum harmala</i>	42.1	18.2	10.7	1
<i>Banisteriopsis caapi</i>	38.8	23.1	4.1	1
Other stimulants ^b	38.8	8.3	3.3	2.5
DXM	29.8	3.3	1.7	1.5
Heroin	26.4	5.8	1.7	11.5
GHB	24.8	1.7	0.8	1

^a Among those who had ever used the drug.

^b Involving non-medical use.

Including alcohol and tobacco, the median number of discrete drugs that had been consumed by the sample in the previous 30 days was 6 (range 0–16).

Hallucinogens most widely consumed included LSD (96.7%), psilocybin mushrooms (91.7%), *S. divinorum* (67.8%), mescaline (64.5%) and ketamine (56.2%). Several plant-based drugs associated with DMT use were also frequently reported by the sample; the MAO inhibitors *Peganum harmala* (Syrian rue; 42.1%) and *B. caapi* (38.8%), as well as the compounds 5-methoxy-dimethyltryptamine (5-MeO-DMT) and bufotenin, which are present in *Anadenanthera peregrina* snuffs (11.6%; Ott, 1996).

Respondents reported use of 'party drugs' such as MDMA (94.2%), meth/amphetamine (83.5%), cocaine (72.7%), amyl nitrite (60.3%) and GHB or related compounds (24.8%). Over one-quarter reported previous use of heroin (26.4%), while past non-medical use of opiates (50.4%), benzodiazepines (58.7%) and stimulants such as methylphenidate (38.8%) was also commonplace.

Several plant-based psychoactives such as kava (*Piper methysticum*; 9.1%), *Amanita muscaria* (6.6%), ergine containing species such as *Ipomoea violacea* and *Argyrea nervosa* (6.6%), opium (6.6%), and anticholinergic deliriantes such as *Datura* spp. and *Brugmansia* spp. (6.6%) were also reported. Interestingly, over 25 synthetic tryptamine and phenethylamine 'research chemicals' (Shulgin and Shulgin, 1991, 1997) were also widely reported. The most commonly used compounds of this class included 2C-B (44.6%), 2C-I (13.2%), 2C-E (9.1%) and 2C-T-7 (7.4%).

More than a quarter (28.1%) reported having injected drugs intravenously for non-medical purposes on at least one occasion, and 7.4% had previously injected drugs with a needle that had been used by another person. Refer to Table 1 for frequency of drug use in the previous 6 months and 30 days, in addition to median number of days used in the past 30 days.

3.3. DMT use

The median age of initiation of DMT use was 24 years (range 15–65 years). The median total number of occasions respondents had used DMT was 10 times (range 1–400 times). The most common route of administering DMT was smoking (98.3%), fol-

lowed by oral administration in the form of ayahuasca (30.6%). Those who reported having previously consumed ayahuasca took DMT on significantly more occasions than those who did not (median=30 versus 6, $U=732$, $p<0.001$). Less common routes of administration included insufflation (5%) and injection (2.5%).

Eleven participants (9.1%) had consumed DMT on only one occasion; 34.7% had used it on 20 or more occasions. Approximately two-thirds (68.6%) of respondents had used DMT in the past 6 months, while 86.7% had used it in the past 12 months. Median duration of DMT use was 2 years (range 6 months to 31 years).

The most frequently cited sources for first having heard about DMT was primarily through friends (47.9%), the Internet (24.8%) and print media (22.3%). The median number of friends that the respondent had stated had also taken DMT was 15 (range 1–500). There was a strong positive correlation between the number of friends that had taken DMT with the respondent's total occasions used ($r=0.67$, $p<0.001$). The median number of people that the respondents had given DMT to was 3 (range 0–200).

3.4. Obtaining DMT

The reported usual sources for obtaining or purchasing DMT included friends (60.3%) and acquaintances (9.1%), friends that were dealers (14.9%), known dealers (5.8%) and rarely, dealers that were strangers (2.5%). Moreover, approximately one-quarter (26.4%) reported that they extract DMT themselves from plant material that they find, grow or purchase. Obtaining or purchasing the necessary plant material was rated as very easy (26.4%), somewhat easy (30.6%), somewhat difficult (20.7%), very difficult (9.1%), and unsure (13.2%). Almost half of respondents (48.8%) reported not knowing the street price of 1 g of ready-to-consume DMT. Of those that did, the median quoted price was \$150/g (range \$35–1000/g). The ease with which DMT could be obtained was rated as very difficult (19.8%), somewhat difficult (36.4%), somewhat easy (26.4%), very easy (11.6%), and unsure (5.8%).

3.5. Reasons for trying

Participants were asked why they initially tried DMT, the reasons for which were subsequently coded according to themes ($n = 103$). The most commonly cited reasons were out of a general interest in hallucinogenic or plant-based drugs (46.6%) and curiosity about DMT's effects (41.7%). Interestingly, almost one-third (31.1%) stated that they tried DMT for its potential psychotherapeutic benefits, such as personal insight or self exploration. Of a similar theme was the potential spiritual benefits of DMT (29.1%) such as "finding God" or "finding spiritual meaning". Being given DMT or having it recommended by others was also mentioned (18.4%) as a reason for trying the drug. Only three (2.9%) respondents stated that they first consumed DMT for fun.

3.6. Location of usage

Respondent's usual places of DMT use were: their own home (81.8%), outdoors (55.4%), a friend's home (52.1%), doofs (24.8%), private parties (16.5%), public places (12.4%) and the home of an acquaintance (5%). Users generally took DMT in a small group of up to four friends (76.9%), alone (52.1%), with their partner (26.4%), in a large group of five or more friends (9.1%) or in ayahuasca ceremonies (5%). Those who reported commonly taking DMT alone used DMT on significantly more occasions than those who did not (median = 20 versus 5, $U = 971$, $p < 0.001$).

3.7. Smoked DMT

All but two of the 121 respondents (98.3%) reported having previously smoked DMT on at least one occasion. The median number of occasions users had smoked DMT was 8 (range 1–400). By far the most common methods for smoking were through a waterpipe or 'bong' (55.5%), glass pipe (54.6%), or in a cigarette or joint (29.4%). Other methods included non-glass pipes (10.1%) and vaporisers (5%).

Approximately two-thirds (68.1%) of DMT smokers reported commonly smoking DMT with other drugs. In those who did, cannabis was the most commonly combined drug (53.1%), followed by LSD (28.4%), alcohol (27.2%), psilocybin (19.8%) and MDMA (8.6%). Moreover, 42% of DMT smokers reported concomitant use of an MAO inhibitor such as *B. caapi* or *P. harmala*.

Respondents were asked to nominate up to five positive ($n = 98$) and five negative aspects ($n = 103$) of smoked DMT and responses were coded according to themes (Table 2). Three-quarters (75.5%) of the sample reported a positive aspect of smoked DMT to be a personally meaningful or insightful experience, and over half (54.1%) cited spiritual experiences. Feelings of euphoria (29.6%), the brief duration of smoked DMT (27.6%) and the intense visual hallucina-

tions produced under its influence (21.4%) were also mentioned as positive effects.

When asked what the negative aspects of smoked DMT were almost half (49.6%) reported difficulties with the act of smoking the drug. This included the bad taste, coughing and possible respiratory damage from smoking. Other negative aspects included psychological or spiritual conflicts (23.5%). Descriptions included an overwhelming change in psychological or spiritual outlook or worldview that may be difficult to integrate once the effects of DMT had worn off. Approximately 1 in 10 (10.9%) also noted an experience under the influence of DMT characterised by marked anxiety or stress. Interestingly, the brief duration of smoked DMT was also cited as a negative aspect of the experience (10.9%), as was its illegality (5.9%).

3.8. Ayahuasca

Thirty-seven (30.6%) respondents reported having consumed ayahuasca on at least one occasion. Interestingly, only two ayahuasca users reported not having smoked DMT. Thus, although many DMT smokers had not consumed ayahuasca, the vast majority (94.6%) of ayahuasca users had also smoked DMT. Analysis indicated that ayahuasca users had smoked DMT on more occasions than those who reported only smoking DMT (median 20 versus 7, $U = 851.5$, $p < 0.001$). The median number of occasions ayahuasca had been consumed was four times (range 1–180).

As previously noted, an MAO inhibitor must be consumed for DMT to be orally active (McKenna, 2004). The most commonly reported MAO inhibitors were *B. caapi* (73%) and *P. harmala* (54.1%). In contrast to smoking DMT, users reported not usually combining ayahuasca with any other drugs (73%). This was generally independent of whether or not the same individual reported smoking DMT with other drugs. In the 10 users who commonly used other drugs with ayahuasca, the most frequently used other drugs were cannabis (7 users) and psilocybin (3).

As with smoked DMT, respondents were asked to cite the positive ($n = 30$) and negative ($n = 25$) aspects of ayahuasca (Table 2). Almost half (46.7%) felt that the ayahuasca experience offered them personal insight. One-third also stated that the experience was more psychologically cleansing or cathartic in comparison to smoked DMT. The longer duration of ayahuasca relative to smoked DMT (30%) and a somewhat smoother onset of action and more gentle experience (30%) were also noted. The unusual psychic phenomena characteristic of ayahuasca such as apparent near-death or out-of-body experiences (Shanon, 2002) was also mentioned as a positive aspect of the preparation (20%).

Ayahuasca's putrid taste and the associated nausea characteristic of the experience were by far the most commonly reported negative aspect (44%). One-fifth (20%) also reported ayahuasca's

Table 2

Five most frequently mentioned positive ($n = 98$) and negative ($n = 103$) aspects of smoked DMT, and positive ($n = 30$) and negative ($n = 25$) aspects of ayahuasca.

Positive aspects	Percentage (%)	Negative aspects	Percentage (%)
Smoked DMT			
Meaningful or insightful	75.5	Bad taste or coughing	49.6
Spiritual experiences	54.1	Psychospiritual distress	23.5
Euphoria	29.6	Anxiety or stress during	10.9
Short duration	27.6	Short duration	10.9
Visual hallucinations	21.4	Legal status	5.9
Ayahuasca			
Meaningful or insightful	46.7	Nausea, taste, or purge	44.0
Cleansing or cathartic	33.3	Long duration	20.0
Long duration	30.0	Psychospiritual distress	16.0
Smooth, gentle experience	30.0	Intensity	12.0
Unusual psychic phenomena	20.0	Poorly prepared brews	12.0

long duration. It was commonly reported that unlike smoked DMT, the longer duration of ayahuasca meant that the trip could not end shortly thereafter should a negative experience occur. As with smoked DMT, psychospiritual conflict was also mentioned (16%) as a negative aspect of ayahuasca. Ayahuasca's intensity (12%) and poorly prepared brews (12%) were also noted as negative aspects.

3.9. Perceived dangers

Participants rated on a 5-point scale how safe they thought it was to use DMT for them personally. The majority of the sample generally considered their use to be safe, with most deeming it to be either very safe (54.5%), quite safe (38%) or moderately safe (6.6%). No participant reported having sought medical attention as a consequence of their DMT use, and only one stated having previously driven whilst under the influence of ayahuasca (2.7% of ayahuasca users).

Respondents were asked to state what they felt were the three main risks about DMT use in general ($n = 111$). The most frequently reported risk for DMT was having a bad trip (50.5%), or a frightening experience while under the influence of DMT. Commonly cited was the potential for psychospiritual problems arising as a consequence of DMT use (39%). Responses generally reflected the belief that DMT could produce an overwhelming change in spiritual or psychological outlook or worldview that may be difficult to integrate once the effects of the drug had worn off. Over a quarter (26.3%) also stated aversive physiological reactions from consuming DMT. In the case of smoked DMT, respiratory irritation from inhaling smoke was the primary complaint while nausea was the main physical concern for ayahuasca use. Other dangers included falls or burns arising as a consequence of not remaining seated under its influence (22.1%) as well as the disrespectful or irreligious use of DMT (17.9%).

When asked how the risks associated with DMT could be reduced ($n = 103$), approximately half (50.5%) mentioned the need for greater education regarding safe use. The presence of another individual was deemed important (34.7%) as well as the need to consume DMT with an appropriate mindset (28.4%) and in a suitable context or environment (28.4%). The legalisation of DMT (32.6%) and limiting one's use of the drug (29.5%) were also mentioned. Ensuring purity of the DMT consumed (9.5%) was also noted as an important factor in harm minimisation.

4. Discussion

This study provides the first comprehensive investigation of recreational DMT use by examining 121 Australian DMT users. Sample demographics indicate that DMT users are generally employed or undertaking higher education, and are less likely to be subject to the social and economic marginalisation characteristic of other drug using populations (Room, 2005).

The main routes of administration for recreational use were smoking DMT and ayahuasca, with smoking being the most common method. Given that almost all ayahuasca users had also smoked DMT, but not all DMT smokers had used ayahuasca, the data suggest that the typical pattern of use commences with smoked DMT. Thereafter, some but not all users will progress to ayahuasca use, but clearly this requires further investigation.

4.1. Pattern of use

4.1.1. Initiation and use. Consumption of DMT appears to be mediated by peers. Friends were the primary source in regard to first hearing about DMT and were a common source for providing the drug. This suggests that a friend will initiate an individual to DMT, who will subsequently introduce it to others. The high correlation

between the number of friends who had taken DMT and the respondent's own times used further supports this. The influence of peers in drug initiation and use is well characterised (Doherty et al., 2000; Andrews et al., 2002; Kandel et al., 1978) and this appears to be the case with DMT.

An interest in hallucinogens and curiosity about the effects of DMT seemed to be the overarching reasons for initiation of use. Although it was not examined in the present study, it would be of benefit to determine the drugs used preceding initiation of DMT use. Of particular relevance was the fact that many sought to consume DMT for psychospiritual benefits. This is consistent with the ritualised use of ayahuasca by members of the UDV and Santo Daime (McKenna, 2004; Shanon, 2002). In our unpublished qualitative data, the belief in the psychotherapeutic or spiritual benefits of the drug appears to be a recurrent factor influencing continued use of DMT.

4.1.2. Smoked DMT. Smoking was the most common route of administration and had been employed by almost all respondents. Smoked DMT is frequently combined with cannabis or other hallucinogens. Anecdotally, DMT users sprinkle DMT over cannabis which is subsequently smoked through a bong (Turner, 1994). Of note is the concomitant use of plant-based MAO inhibitors such as *B. caapi*, which would presumably prolong the duration of smoked DMT by inhibiting its metabolism (McKenna, 2004). The co-administration of these plants may suggest a purposeful attempt to extend the duration of the experience. However, the motivations underpinning concomitant use of MAO inhibitors requires further investigation before any stringent conclusions can be drawn.

4.1.3. Ayahuasca. Ayahuasca was less commonly used, having been consumed by 30.6% of the sample. As previously noted, DMT's short duration when smoked was cited as a negative aspect of the drug, and it is possible that smokers who subsequently progress to ayahuasca use do so from a desire for a prolonged drug effect. It is also of note that ayahuasca users had smoked DMT on more occasions than those who had not. This may indicate that in a subset of DMT users, individuals who develop a predilection for the drug may then progress to ayahuasca use. This, however, remains entirely speculative.

Aside from the necessary use of MAO inhibitors to make DMT orally active, users of ayahuasca generally do not mix it with other drugs. This is an important departure from smoked DMT. The use of MAO inhibitors in conjunction with other drugs interacting with monoamine neurotransmitters such as antidepressants may result in potentially fatal serotonin syndrome (Callaway and Grob, 1998). Owing to the intensity of the ayahuasca experience (Shanon, 2002), the absence of concomitant drug use among those consuming the preparation may reflect an aversion towards intensifying what may already be an overwhelming experience, or perhaps an unwillingness to risk dangerous drug interactions associated with MAO inhibitor use. Investigating users' awareness of such risks may warrant attention in future studies, as it is arguably the most pressing health-related risk associated with ayahuasca use.

4.1.4. Positive and negative aspects. Regardless of the route of administration, respondents cited an increase in personal or spiritual insight as a positive aspect of their DMT use. This is consistent with reports of ayahuasca users in formal religious groups such as the UDV and Santo Daime (Shanon, 2002; Grob et al., 1996). Naturally occurring hallucinogens such as ayahuasca have a long history of religious use, and the recent rise in neo-shamanism reflects a resurgence of Western interest in these compounds (Tupper, 2008, 2009). For instance, psilocybin has been found to produce religious experience in healthy volunteers (Griffiths et al., 2006).

Of further interest is the antipodal attitude towards drug duration. For both DMT smokers and ayahuasca users, the short and the long duration of each route respectively was perceived as both a good and bad aspect of the drug depending on the respondent. For example, while some were amenable to the short duration of smoked DMT, others considered this to be a shortcoming. Similar considerations can be made of ayahuasca. The degree to which duration of effects influences patterns of DMT use requires further investigation, although ayahuasca users appeared to use this preparation less frequently, possibly due to the nausea also associated with such use (Shanon, 2002).

The most commonly cited negative aspect of both routes of administration was unique to the route of administration. Smoked DMT causes coughing and is oftentimes described as having a harsh, plastic-like taste (Turner, 1994), while ayahuasca commonly produces nausea and emesis after administration (Shanon, 2002). Among traditional users of ayahuasca, emesis or '*la purga*' is said to be both cathartic and restorative. It remains entirely possible that this unpleasant experience of nausea may be a contributing factor to the 'cleansing' aspects reported regarding ayahuasca usage in this sample. Psychospiritual conflict is a commonly cited negative aspect of the DMT experience, regardless of the method of administration. This is discussed in greater detail below.

4.1.5. Perceived dangers and risks. The majority of DMT users recruited for this study believed their use of DMT to be safe. Although previous studies examining supervised religious use of ayahuasca found associated harms to be negligible (e.g. Halpern et al., 2008), a level of concern may be directed at the finding that approximately half of the sample commonly consume DMT alone. This is despite general acknowledgement that consuming DMT in the company of others would minimise the risks of an adverse drug effect.

Moreover, the safety profile of illicit DMT remains unclear. While pure DMT and ayahuasca are well-tolerated (Strassman, 2001; Gable, 2007) the majority of DMT available in Australia appears to be extracted from plant matter by individuals typically lacking any formal qualification in organic chemistry (Tramacchi, 2006). Aside from DMT, the presence of other psychoactive tryptamines such as 5-MeO-DMT in the relevant plant matter (Ott, 1996) suggests that it is unlikely that DMT users consume the pure compound, but rather a mixture of DMT and related analogues (Tramacchi, 2006). This raises potentially serious questions as to the safety and purity of illicit DMT, a point widely acknowledged by the present sample. Other, more obvious physiological harms such as respiratory damage from smoking are acknowledged by users, although the extent to which such risks influence patterns of DMT use remains unclear.

In addition to physical harm, the present sample perceived psychological trauma as a risk of DMT use. Hallucinogens in general have been reported as potential catalysts of rapid and profound changes in psychological or spiritual outlook (Grof, 2008; Hofmann, 2005), and although such changes may be beneficial for some users, in others they may produce considerable distress. When used in an unsupportive environment the likelihood of negative psychological impacts from many hallucinogens increases, an observation readily noted in the present sample. The perceived risks from DMT are predominantly psychological in nature, from the frightening experience of acute intoxication – a 'bad trip' – to psychospiritual disturbances that may accompany and outlast the primary drug effects. Although we did not explicitly examine the frequency with which this occurred in the sample, this is an important question for future research.

The need to remain informed about the risks inherent in DMT use was the most frequently cited harm minimisation strategy. Common responses about harm minimisation also centred around

using DMT in an appropriate "set and setting" – the physical and social environment in which a drug is consumed as well as the subject's emotional state, intentions, expectancies and beliefs concerning the use of a psychoactive substance (Faillace and Szára, 1968). The practise of this would also minimise the possibility of falls or burns that may occur due to DMT intoxication. The addition of a 'sitter', or another individual who may render assistance if required, was also stated as a factor in using the drug safely.

4.2. Pattern of other drug use

In this sample DMT is part of an extensive pattern of drug use involving a diverse class of drugs. In addition to substances such as alcohol, cannabis, psychostimulants, depressants and narcotics, the use of a wide array of otherwise obscure drugs was reported. These drugs included plant-based psychoactives as well as novel tryptamine and phenethylamine-based hallucinogens described by Shulgin and Shulgin (1991, 1997). Although multiple drug use is commonplace among drug users (Wilkinson et al., 1987), the degree to which this was observed in the present sample is exceptional. For example, in his study of 135 Scottish MDMA users, Forsyth (1996) found that 51 discrete drugs had been used by respondents; in the present study, nearly double that number (92) was reported. Moreover, while Forsyth found the mean number of discrete drugs used to be 10.7, in this sample this was 17.3. Although there appears a need to raise concern within our sample, this level of concern is perhaps best directed at the history of drug use, rather than with DMT use per se. For example, the use of cannabis seems ubiquitous, and approximately a quarter of the sample may be cannabis dependent. In addition, over one-quarter of the sample reported lifetime use of intravenous drugs. Thus, although hallucinogens appeared to be favoured by the sample, to describe the sample primarily as DMT or hallucinogen users would be inaccurate. Future studies would benefit from determining the degree to which users favour DMT relative to other drugs they consume.

5. Conclusions

This sample of recreational DMT users was recruited primarily by snowball sampling. Snowballing has the advantage of using social networks to gain access to 'hidden' populations such as illicit drug users (Thompson and Collins, 2002; Biernacki and Waldorf, 1981). This is particularly advantageous given that DMT remains an infrequently used drug in most Western countries including Australia, and it is doubtful that such a large sample could be obtained without doing so. However, the very benefits of snowball sampling may lead to potential problems in sample biasing. For instance, the majority of respondents in the sample were male and it remains unclear whether the data are a true representation of Australian DMT users in general, or a bias in the sample.

It is also necessary to note that aside from containing DMT, plant extractions of DMT are likely to comprise other psychoactive tryptamines such as 5-MeO-DMT and bufotenin (Ott, 1996). Thus, to consider this to be a study of DMT use exclusively is akin to presuming that ecstasy use constitutes the use of unadulterated MDMA. In the absence of forensic data of seized DMT, the exact purity and chemical constituents of material purported to be DMT remains unknown.

Despite evidence that the use of DMT in Australia and other Western countries is gaining in popularity (e.g. Grigoriadis, 2006; Cox, 2009; Greenhouse, 2006), it is prudent to note that DMT presently remains an uncommon drug. There are several reasons for this. First, as is generally the case with other hallucinogens, DMT is neither physically nor psychologically addictive (Callaway et al.,

1999; Strassman, 2001). Moreover, many are drawn to DMT out of curiosity, yet the very effects to which users are attracted are also likely to moderate its frequency of use. Its intense and unpredictable effects, and individual susceptibility to having a frightening experience or bad trip (Strassman, 2001) are likely to dissuade many recreational users from consuming it on more than one occasion.

Although largely exploratory, this study offers a framework within which to approach future studies of recreational DMT use. First, it demonstrates that smoking is the most common route of administration for recreational users, whereas studies previous to this have predominantly investigated ayahuasca use (e.g. Grob et al., 1996; Shanon, 2002; Trichter et al., 2009). Although recruiting DMT users through formal religions such as the UDV and Santo Daime offers a readily accessible sample of DMT users, future studies may benefit from examining DMT use beyond these institutions. Even beyond these churches, 'recreational' DMT use appears to reflect an earnest attempt at gaining psychospiritual insight through means of altered states of consciousness. Future studies should seek to examine the psychological characteristics of recreational DMT users, their attitudes and motivations, and actual benefits and harms, if any, associated with DMT use.

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V.C. and J.P. developed the questionnaire and recruited participants. A.M. conducted data analysis and literature review. V.C. wrote the first draft of the manuscript, and J.P. and A.M. subsequently edited and finalised drafts. All authors contributed to and have approved the final manuscript.

Conflict of interest

No conflict declared.

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